Requirements Specification for Energy Production Monitoring Business Process

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1. General description of business process

a. Process Overview & Performance Metrics

The process of energy production monitoring involves continuously capturing power generation data from individual solar panels and wind turbines. The smart meters installed on each unit read the energy generated in kWh and transmit this data to the central monitoring system (PowerMaster). The system logs the production amount, timestamp, and operational status of each unit.

Measurable goals:

- Increase the average monthly energy output by at least 1% compared to the same month in the previous year.
- Reduce the number of days where total daily energy production falls below 80% of the average daily production for the corresponding month in the previous year by 10%.

b. Typical Analytical Questions

- Compare the energy produced by wind turbines versus solar panels this month.
- What are the most efficient turbines in this quarter?
- Calculate the average daily energy production for the last year.
- Compare the energy production of the current month with the production of the previous month.
- What is the impact of wind speed on energy output?

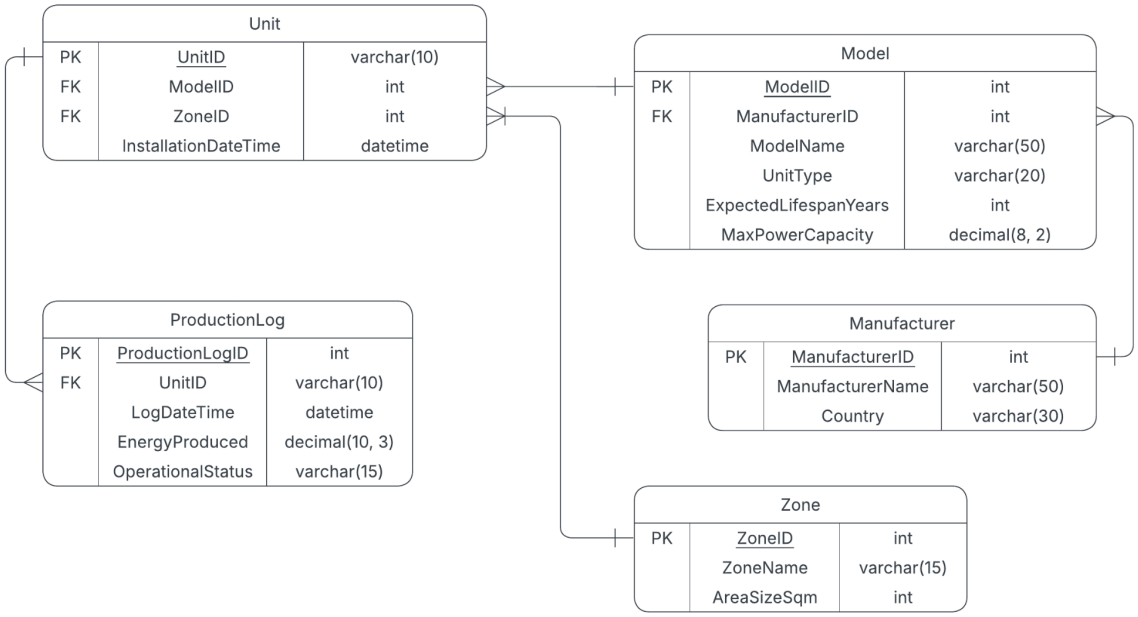
c. Data

All production data is extracted from the relational system named "PowerMaster". It stores information about the generation units, their locations, and the production logs (timestamps and kWh produced).

In addition, meteorological data (temperature, wind speed, weather conditions) is provided hourly by a Polish meteorological institute and stored in a CSV file format. This data can be integrated with the production logs using dates/timestamps and ZoneIDs.

2. Data sources structures

PowerMaster (Relational Database)



Manufacturer

A company that produces energy generation equipment.

ATTRIBUTE	TYPE	DESCRIPTION
ManufacturerID	Integer	PK, auto-generated ID for the manufacturer.

ATTRIBUTE	TYPE	DESCRIPTION
ManufacturerName	Varchar(50)	Name of the manufacturing company.
Country	Varchar(30)	Country of origin of the manufacturer.

Model

A specific type of equipment produced by a manufacturer, determining its technical capabilities and type. There are 2 types of units, solar panels and wind turbines.

ATTRIBUTE	TYPE	DESCRIPTION
ModelID	Integer	PK, auto-generated ID for the equipment model.
ManufacturerID	Integer	FK pointing to the Manufacturer table.
ModelName	Varchar(50)	The specific model name of the equipment.
UnitType	Varchar(20)	Type of unit ('Solar' / 'Wind').
ExpectedLifespanYears	Integer	The manufacturer's stated lifespan in years.
MaxPowerCapacity	Decimal(8, 2)	Maximum theoretical output capacity in kW.

Zone

A geographical sector of the energy farm where generation units are grouped.

- Bydgoszcz
- Gdansk
- Warsaw
- Olsztyn
- Szczecin
- Zielona Gora
- Bialystok

- Poznan

ATTRIBUTE	TYPE	DESCRIPTION
ZoneID	Integer	PK, auto-generated identifier for the farm zone.
ZoneName	Varchar(15)	The specific name of the zone (e.g., 'Bydgoszcz').
AreaSizeSqm	Integer	The size of the zone in square meters.

Unit

A single, physical energy-generating asset installed at a specific location on the farm.

ATTRIBUTE	TYPE	DESCRIPTION
UnitID	Varchar(10)	PK, unique identifier of the generation unit.
ModelID	Integer	FK pointing to the Model table.
ZoneID	Integer	FK pointing to the Zone table.
InstallationDateTime	DateTime	The date the unit was installed.

ProductionLog

A record generated by a unit's smart meter, logging the exact amount of energy produced at a specific timestamp.

ATTRIBUTE	TYPE	DESCRIPTION
ProductionLogID	Integer	PK, auto-generated incrementing ID.
UnitID	Varchar(10)	FK pointing to the Unit table.
LogDateTime	DateTime	Date and time of the production reading.
EnergyProduced	Decimal(10, 3)	Energy produced in kWh.

ATTRIBUTE	TYPE	DESCRIPTION
OperationalStatus	Varchar(15)	Status of the unit at the time of the reading(‘Operating’ / ‘Fault’ / ‘Maintenance’).

Weather_Data (CSV File)

One row per hour per zone. Column headers in row 1.

- **Column A:** Datetime (ISO 8601 format: yyyy-MM-ddTHH:mm:ss)
- **Column B:** ZoneID (numeric, matches ZoneID in the Zone table)
- **Column C:** AvgTemperature (numeric, Celsius)
- **Column D:** AvgWindSpeed (numeric, m/s)
- **Column E:** Precipitation (numeric, mm)
- **Column F:** CloudCover (numeric, integer in range 0-8)
- **Column G:** SunshineDuration (numeric, minutes of sunshine per hour)

3. Scenarios of analytical problems

Problem 1: Why did energy production fluctuate this month?

1. Compare the total energy produced in the current month with the previous month.
2. What are the top 5 generating units in the current month?
3. Compare the average daily energy production of solar units on 'Sunny' (CloudCover < 2) days versus 'Cloudy' (CloudCover > 6) days.
4. What is the average daily energy produced by wind turbines across different wind speed ranges (0-3 m/s, 3-6 m/s, 6-9 m/s, >9 m/s)?
5. Compare the total energy production across all distinct zones in the last quarter.

Problem 2: What is the impact of equipment aging on performance under various conditions?

1. Compare the energy produced by units which are over 5 years old with those installed recently (meaning less than 1 year ago).
2. What is the average energy produced by older units (installed > 5 years ago) when the average temperature exceeds 30 degrees Celsius?

3. Which unit type ('Solar' or 'Wind') shows a steeper decline in energy production as it ages?
4. What are the lowest producing units on days with optimal weather conditions (wind speed >9 m/s wind for wind units, CloudCover < 2 for solar units)?
5. Compare the monthly production of the oldest wind turbine in the current month relative to the same month last year.

4. Data needed for analytical problems

Problem 1 Mapping: Energy Production Fluctuation

1. **Compare the total energy produced in the current month with the previous month.**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Month: PowerMaster, table ProductionLog, column LogDateTime
2. **What are the top 5 generating units in the current month?**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Unit: PowerMaster, table ProductionLog, column UnitID
    - Month: PowerMaster, table ProductionLog, column LogDateTime
3. **Compare the average daily energy production of solar units on 'Sunny' days versus 'Cloudy' days.**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Unit Type: PowerMaster, table Model, column UnitType ('Solar') (Join required from Unit to Model)
    - Date and Zone mapping: PowerMaster, tables ProductionLog, Unit, and Zone (join required from Unit to Zone using ZoneID to get the ZoneName)
    - Cloud cover: Weather_Data CSV, column F (CloudCover)
    - Sunshine Duration: Weather_Data CSV, column G (SunshineDuration)
4. **What is the average daily energy produced by wind turbines across different wind speed ranges (0-3 m/s, 3-6 m/s, 6-9 m/s, >9 m/s)?**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced

    - Unit Type: PowerMaster, table Model, column UnitType ('Wind') (Join required from Unit to Model)
    - Date mapping: PowerMaster, table ProductionLog, column LogDateTime
    - Wind speed: Weather_Data CSV, column D (AvgWindSpeed)
5. **Compare the total energy production across all distinct zones in the last quarter.**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Zone: PowerMaster, table Zone, column ZoneName (Joined through Unit table using ZoneID)
    - Date: PowerMaster, table ProductionLog, column LogDateTime

Problem 2 Mapping: Impact of Equipment Aging

6. Compare the energy produced by units which are over 5 years old with those installed less than 1 year ago**.**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Installation Date: PowerMaster, table Unit, column InstallationDateTime
7. **What is the average energy produced by older units (installed > 5 years ago) when the average temperature exceeds 30 degrees Celsius?**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Installation Date: PowerMaster, table Unit, column InstallationDateTime
    - Temperature: Weather_Data CSV, column C (AvgTemperature > 30)
    - Date and Zone mapping: PowerMaster, tables ProductionLog, Unit, and Zone
8. **Which unit type ('Solar' or 'Wind') shows a steeper decline in energy production as it ages?**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Unit Type: PowerMaster, table Model, column UnitType (Join required from Unit to Model)
    - Installation Date: PowerMaster, table Unit, column InstallationDateTime
    - Date of Production: PowerMaster, table ProductionLog, column LogDateTime

9. **What are the lowest producing units on days with optimal weather conditions (wind speed >9 m/s wind for wind units, CloudCover < 2 for solar units)?**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Unit Type: PowerMaster, table Model, column UnitType (Join required from Unit to Model)
    - Unit details (Manufacturer/Model): PowerMaster, tables Manufacturer and Model (joined through Unit)
    - Weather condition: Weather_Data CSV, columns D and F
    - Date mapping: PowerMaster, table ProductionLog
10. **Compare the monthly production of the oldest wind turbine in the current month relative to the same month last year.**
    - Energy produced: PowerMaster, table ProductionLog, column EnergyProduced
    - Unit Type: PowerMaster, table Model, column UnitType ('Wind') (Join required from Unit to Model)
    - Installation Date: PowerMaster, table Unit, column InstallationDateTime (to find the minimum date)
    - Month: PowerMaster, table ProductionLog, column LogDateTime

Queries Requiring Additional Data / Changes

- **Query demanding additional data sources (without changing the business process):**
  "Is the recent drop in production correlated with high levels of atmospheric dust/smog?"
    - **Data needed:** Air quality/dust concentration data. This can be obtained from external environmental APIs or public governmental sensors without altering the farm's internal processes.
- **Query demanding additional data (requiring changes in the business process):**
  "Does the time since the last panel cleaning affect the daily energy yield of solar units?"
    - **Data needed:** Dates of panel cleaning for each solar unit. Currently, cleaning is done ad hoc and is not recorded anywhere. Only failures are logged, and only in the maintenance system. To answer this question, the business process must be changed: the cleaning crew must register every

cleaning (UnitID, date, method) after it’s done, for example as a new “Cleaning” ticket type in the maintenance system. Only then can the days since the last cleaning be calculated for every production record.